



Experimental study on pulse self-heating of lithium-ion battery at low temperature

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ABSTRACT

Battery warming at low temperature is a critical issue affecting battery thermal management. In this study, the pulse self-heating strategy is proposed to enable quick and safe warming of lithium-ion battery at low temperature. The battery is heated up using pulse self-discharge. This strategy can heat up 18,650 commercial battery with a control circuit and alleviate the battery degradation during heating. Compared with continuous direct current self-heating, the battery can be heated up from $-10\text{ }^{\circ}\text{C}$ to $10\text{ }^{\circ}\text{C}$ by pulse heating within 175 s while the direct current heating consumes 280 s with approximating polarization voltage. The effects of ambient temperature, switching interval, and initial state-of-charge on heating performance are further investigated. Heating duration is found to be insensitive to the switching interval in the studied range. Conversely, battery internal resistance and off-period voltage at different ambient temperatures and initial state-of-charges are found to be the predominant factors that influence heating duration. Heating duration in the range of 150–180 s is achieved at ambient temperature ranging from $-10\text{ }^{\circ}\text{C}$ to $0\text{ }^{\circ}\text{C}$. In real applications, pulse heating is recommended to be performed at higher initial state-of-charge (>0.8), at which heating duration is within 200 s.

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1. Introduction

Over the last two decades, lithium-ion batteries (LIBs) have become important power sources for portable devices, such as power tools, laptops and smart phones. Next-generation LIBs are further utilized in large-scale applications, such as electric vehicles, power grids, and energy storage. Thermal management is a critical issue in LIB applications. Prior to designing proper thermal management methods, the heat generation and temperature distribution in the single battery and battery pack should be clarified. To solve this problem, Panchal et al. [1,2] have conducted numerical and experimental investigations on the battery thermal behavior and surface temperature distribution. They presented the surface temperature distribution pattern for large format pouch cells. The battery cooling methods have been widely discussed. Various cooling methods including air cooling [3], liquid cooling [4] and phase change material cooling [5,6] have been proposed. However, the investigations on battery heating strategies at low temperature are insufficient.

LIBs suffer from poor performance at low temperature, at which a slowdown of specie transfer and chemical reactions fundamen-

tally occur. As stated by Zhang et al. [7], the conductivity of electrolyte composed of LiPF₆, propylene carbonate, ethylene carbonate and ethyl methyl carbonate dropped from $1.3 \times 10^{-2}\text{ S/cm}$ to $1.0 \times 10^{-3}\text{ S/cm}$ when temperature decreased from $20\text{ }^{\circ}\text{C}$ to $-40\text{ }^{\circ}\text{C}$. The ion diffusivity in anode suffered a significant decrease when temperature dropped from $20\text{ }^{\circ}\text{C}$ to $-20\text{ }^{\circ}\text{C}$ [8]. The decrement in electrolyte conductivity and lithium diffusivity significantly increased internal resistance of the battery. As indicated by Zhang et al. [9], the cell resistance increased from $0.1\text{ }\Omega$ to $1\text{ }\Omega$ with cell temperature decreasing from $20\text{ }^{\circ}\text{C}$ to $-20\text{ }^{\circ}\text{C}$. These limitations reduce the available energy, power and capacity of a cell. As stated in the study of Huang et al. [10], the battery capacity at $-40\text{ }^{\circ}\text{C}$ was 12% of that at room temperature. More severely, the limited transport kinetics causes the lithium deposit on the anode surface during charging. As a result, lithium plating occurs, which reduces battery capacity [11]. Moreover, the dendrite caused by plating may induce an internal short circuit in the battery and may cause safety issues, such as thermal runaway.

To overcome the performance degradation in LIBs at low temperatures, the batteries should be warmed up prior to use to maintain a normal start up and sufficient energy output. Therefore, effective heating strategies are necessary. The existing heating strategies can be categorized as external heating and internal

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Nomenclature

A	surface area, cm ²
C	discharge capacity, A h
C_p	specific heat capacity of the battery, J/g·K
h	heat transfer coefficient, W/m ² ·K
I	current, A
Li	lithium
m	mass of the battery, kg
N	number of discharging periods
Nu	Nusselt number
Pr	Prandtl number
q	heat generation rate, W
Q	residual capacity, A h
R	resistance, Ω
Ra	Rayleigh number
t	time, s
U	voltage, V
ΔV	voltage drop, V

Greek symbol

η	efficiency
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Subscripts

<i>actual</i>	actual discharge capacity
<i>dis</i>	discharge
<i>DV</i>	discharge voltage

E	external circuit resistance
<i>end</i>	end of heating
h	heating process
I	internal resistance
i	i th discharging period
<i>initial</i>	initial discharge capacity
<i>on</i>	on-period of pulse operation
<i>off</i>	off-period of pulse operation
0	initial time

Acronyms

AC	alternating current
CC	energy consumption
DC	direct current
DCR	direct current resistance
EIS	electrochemical impedance spectra
ECR	energy consumption ratio
IO	input and output
LIB	lithium ion battery
MOSFET	metal–oxide–semiconductor field effect transistor
PTC	positive temperature coefficient
SOC	state of charge
SOH	state of health
SHLB	self-heating lithium ion battery structure
TE	total energy

heating. In external heating strategies, the heating power is transferred to the battery from exterior heat sources. The heat source can be a heating blanket warping the battery or hot fluid blowing the battery. Li et al. [12] adopted heating aluminum plate twined by positive temperature coefficient (PTC) resistance wire to heat up batteries, and they demonstrated that this method improved battery discharge capacity from 21.9 A h to 24.11 A h at low temperature. Pesaran et al. [13] studied the battery heating using external air convection and reported that the battery can be heated up from -40°C to -20°C within 300 s with a 4.71 W h energy input. Song et al. [14] adopted an external air-heating method for battery heating at low temperature and found that a 3.1% reduction in battery capacity occurred when no pre-heating was performed. However, external heating is inefficient due to energy loss into the environment through heat conduction. In addition, non-uniform temperature distribution can occur in the battery pack, thereby inducing degradation in the electrode material and reduction of battery cycle life.

In internal heating strategies, the heating power is generated inside the battery. The easiest way to produce internal heat is to utilize the joule heat generated on the battery internal resistance. Internal heating eliminates the significant heat loss from heat conduction and avoids the heat accumulation in heating devices. Therefore, internal heating method is capable to achieve battery heating with higher efficiency and less thermal-induced battery aging.

The heating source for internal heating strategies can be obtained through two approaches: external power source and self-power of heated battery. In the strategies using external power source, the alternating current (AC) sources is frequently employed and discussed. Zhang et al. [15] proposed a heating method using sinusoidal AC. They discovered that heating rate increases with higher amplitude and lower frequency, this is induced by the higher polarization under high current amplitude and low frequency. Ruan et al. [16] discussed the optimum alternating frequency for AC heating of the battery. They found that

the battery can be heated from -15.4°C to 5.6°C within 338 s at the overall constant frequency of 1377 Hz. This optimum frequency is obtained with optimum polarization voltage and battery impedance. Zhu et al. [17] investigated the effects of current frequencies, amplitudes, and voltage limitations on cell temperature during AC heating process. They found that the AC heating method does not aggravate the cell degradation with the proper voltage protect limitation. In AC heating, the damage caused by continuous discharging or charging can be omitted by high frequency change of the current direction. However, the AC heating is highly restricted to the availability of exterior power source. When the exterior power source is absent, heating cannot be initiated. Therefore, heating strategies that use self-power of heated battery are expected. To fulfil this “standalone” heating mode, the easiest way is to perform the constant current discharge with high current rate (C-rate). The C-rate is a measure of the rate at which a battery is charged or discharged relative to its nominal capacity. Even though, the battery can be warmed up with significant internal heat release, the battery suffers obvious capacity loss with high C-rate discharge. As reported in previous study, the severe battery degradation, 16.9% capacity loss, was observed at 3C discharge rate within 300 cycles for 18,650 LiCoO₂ battery [18]. Ji and Wang [19] discussed the heating of batteries using external direct current (DC) source, and they pointed out high rate bidirectional DC increase the risk of battery degradation. Therefore, both the amplitude and the duration of the current should be restricted to low values in DC heating to avoid Li plating and battery degradation [18,20]. Wang et al. [21] proposed a self-heating lithium-ion battery structure (SHLB) by implanting in a nickel foil with low electric resistance between electrodes within a battery. The battery with the new structure can be self-heated by joule heat generated on the internal nickel foil within 10 s from -30°C to 0°C . Yang et al. [22,23] further compared the battery self-heating method employing SHLB structure with the external heating method. They found that the SHLB is demonstrated to reach a heating speed of $1\text{--}2^\circ\text{C}/\text{sec}$ and the temperature distribution is much spatially

uniform. Lei et al. [24,25] established a numerical model for SHLB to investigate the temperature uniformity inside the prismatic battery. It is found that temperature uniformity in the battery can be increased with low heating power and small battery thickness. However, the structure modification of the battery is technically challenging, and the implantation of nickel can lead to safety issues, such as internal short circuit.

As discussed above, the utilization of heating methods using external DC and AC sources is undesired due to the limited availability of external power source. Moreover, the DC heating would cause electrode degradation at low temperature. Therefore, a self-heating scheme, which eliminates continuous direct discharging/charging process and avoids battery-structure modification, must be exploited. In this study, a pulse internal self-heating strategy is proposed to achieve fast heating of LIB. An electric circuit is built to generate intermittent high current in the battery. The temperature evolutions of batteries under intermittent high current discharge are tested in an environment chamber. The effects of various operation parameters including environmental temperature, pulse switching frequency, and the initial state-of-charge (SOC) on the heating performance are investigated experimentally. The pulse discharge performance at low temperature has not been discussed in previous studies. The present study aims to reveal the pulse discharge characteristics of the lithium ion battery at subzero temperature. The proposed self-heating method is hopeful to provide fast and safe battery warming up with easy design and low cost. Moreover, the results in this study can provide inspiration for new designs in lithium ion battery thermal management system.

2. Experimental

2.1. Design of pulse self-heating scheme

In this study, a pulse signal is used to fulfill the highly efficient self-heating and to eliminate the potential damage in DC self-heating. Commercial 18,650 cells are used to test the proposed self-heating method. The capacity of the battery is 2000 mA h, and the nominal voltage is 3.7 V. The detailed properties of the battery are listed in Table 1. To initiate the intermittent pulse discharge, a circuit module is designed to generate pulse high current in the battery. The design of the discharge circuit is inspired by the test stand design in the investigation of Chen and Sitzman [26], in which the electric load for the test stand is consists of metal-oxide-semiconductor field effect transistor (MOSFET), copper wires and fuses. The test stand is utilized to generate the high pulsed discharge current in the battery at room temperature. In this study, the test stand design is employed to perform the battery self-heating at subzero temperature. As the pulse discharge initiated by MOSFET element easily induces the uncontrollable external short circuit of the battery, an extra variable inductor is added in our design to adjust the load resistance and control the discharge current within the safety limit. Fig. 1(a) shows the schematic of the discharge circuit. The circuit consists of a triode

amplifier transistor, a MOSFET, a hall element, a variable inductor and a controller. In pulse heating, the battery is connected to the circuit. Then, the triode amplifier and MOSFET follow the instruction of the controller to open and close intermittently. The MOSFET has an on-state resistance of roughly 50 mΩ and is capable of holding off 100 V and conducting 200 A at 25 °C. The generated current in the circuit is measured using Hall Effect sensor. The pulse discharging current is produced by the switching process of transistors. The switching interval of transistor's opening and closing can be regulated by MOSFET. Then, a square wave is defined to control the opening and closing of MOSFET. As shown in Fig. 1(b), the circuit is switched on during t_{on} and switched off during t_{off} in one working cycle of the square wave. Therefore, the lower voltage measured during on-period t_{on} is defined as U_{on} and the higher voltage measured during off-period t_{off} is defined as U_{off} . The off-period voltage is lower than the open circuit voltage due to the short heating interval time. The switching on time t_{on} is designed to be equal to the switching off time t_{off} . The value of switching interval can be set by the controller before each group of tests. To avoid thermal run away of the battery at high C-rate, the variable inductor is adjusted to maintain the discharge current at nearly 7.5 C-rate (15 A) during switch-on period of MOSFET at 25 °C. Then, the discharge current should be lower than 15 A at subzero temperature due to increased battery resistance.

2.2. Experiment apparatus

Fig. 2 shows the experimental apparatus. The testing battery is placed in the environment chamber (Nanjing Test Instrument; temperature range: -40 to 150 °C; temperature fluctuation ± 0.5 °C). The battery is connected to the self-designed circuit with a wire to trigger pulse self-heating. The self-designed circuit is located outside the environment chamber. The T-type thermal couple (Omega Engineering Inc., USA) with an accuracy of ± 0.5 °C is used to track the battery surface temperature. To simplify the experiment measurement, one thermal couple is utilized and placed at the center of the battery cylindrical surface. The measurement of the temperature with one thermocouple is conducted based on the fact that the temperature difference at 18,650 battery surface is insignificant. As shown in the study of Ruan et al. [16], five thermocouples are utilized to measure the surface temperature of 18,650 battery with internal heating. They proved that the temperature difference between maximum temperature and average temperature at the battery surface is less than 0.5 °C at 16 A current. The maximum temperature appeared at the battery center. Therefore, the measured temperature at battery center is accurate enough to represent the average temperature of the battery. The temperature and voltage of the battery are recorded with a data acquisition instrument (Agilent 34972, USA, accuracy 0.004%) during heating. The data in acquisition instrument are processed with IO Libraries Suite (Keysight Technology, USA). Prior to each group of tests, the testing battery is placed in the environment chamber for 2 h to ensure the setting temperature. To evaluate the self-heating method, the battery discharging is performed at different switching intervals (0.1, 0.5, and 1 s) and four ambient temperatures (-20, -10, -5, and 0 °C). Restricted to temperature range of the environment chamber, the lowest temperature is set to -20 °C. The four temperature points can provide enough information on the influence of temperature drop. The initial states of batteries are set to different SOC (0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0). Heating is switched off when the temperature increment of the battery reached 20 °C from the initial ambient temperature. Heating duration t_h is defined as the time period from the beginning to the end of heating.

The testing battery is regulated to different SOC using charge-discharge device (Neware BTS3000, China) with an accuracy of

Table 1
Battery properties.

Item	Specification	Manufacturer
Battery	Rated capacity: 2000 mA h Mass: 45 g Surface Area: 36.7 cm ² Specific heat capacity, C_p : 1.05 J/g·K Cathode: LiCoO ₂ Anode: Graphite Electrolyte: LiPF ₆ in organic solvent	Oahel

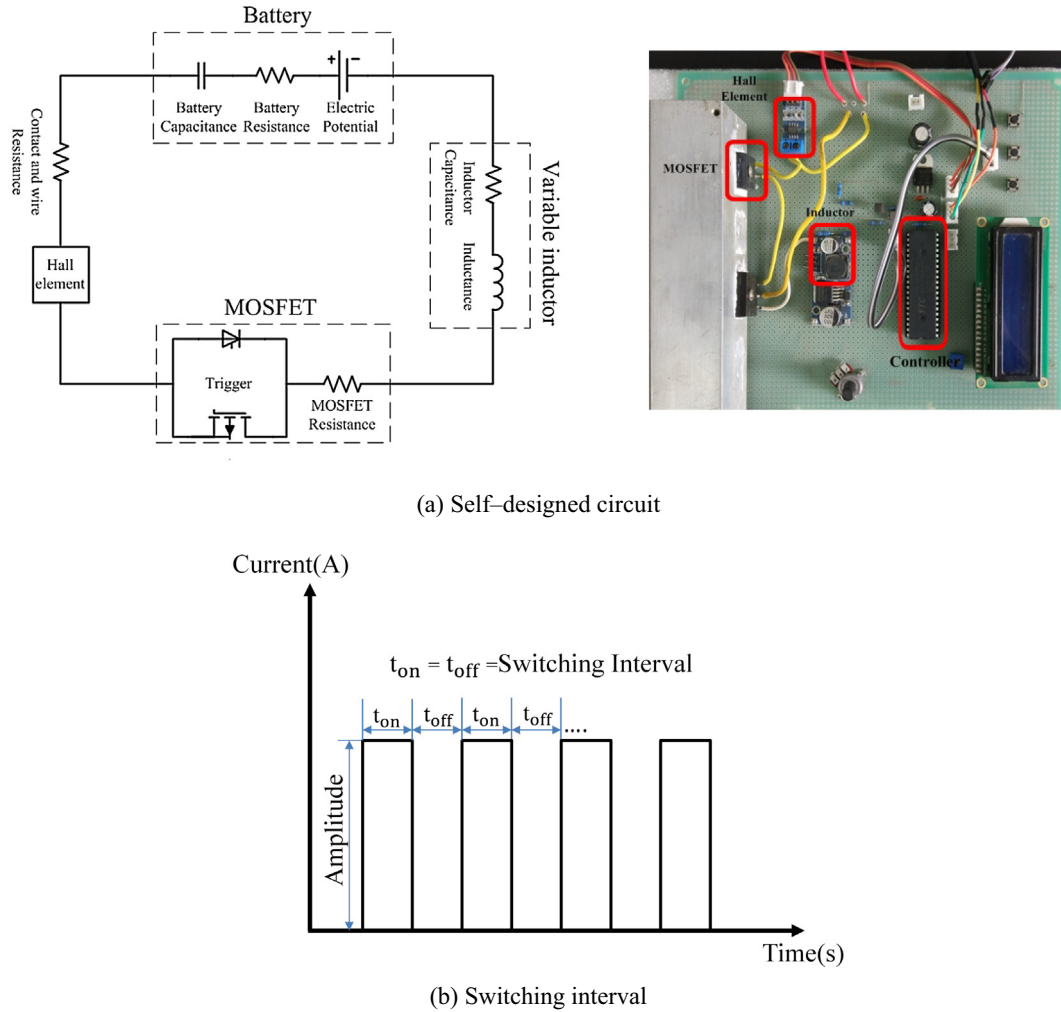


Fig. 1. Self-heating circuit.

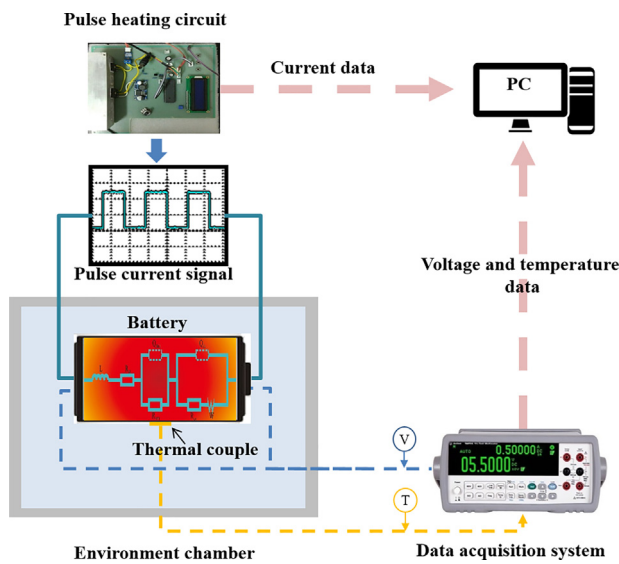


Fig. 2. Experimental apparatus of battery self-heating system.

0.1% at constant current mode. The charge upper limit voltage is set to 4.2 V, the SOC of the LiCoO₂ battery is set to 1 when it is charged to 4.2 V. The cut-off voltage is set to 2.0 V. As shown in

Eq. (1), the ampere-hour counting approach is used to calculate the SOC because it can keep track of the accurate SOC. The SOC is assumed to be 1 when the battery is fully charged to its prescribed voltage (4.2 V).

$$\begin{aligned} \text{SOC}(t) &= \frac{Q(t_0) - \int_0^t I \eta_0 dt}{Q_0} \times 100\% \\ &= \text{SOC}(t_0) - \frac{\int_0^t I \eta_0 dt}{Q_0} \times 100\% \end{aligned} \quad (1)$$

where the $\text{SOC}(t)$ is the SOC at a given time t ; $Q(t_0)$ is the residual capacity at initial time t_0 ; I is the instantaneous current, and the I is a positive value for the discharging process and negative value for the charging process; η_0 is the charge/discharge efficiency and is measured to be 99.5% using charge-discharge device; Q_0 is the rated capacity; and $\text{SOC}(t_0)$ is the SOC at initial time t_0 . For the evaluation of SOC, the discharge process uses constant current mode and the charge process uses constant current-constant voltage mode. To regulate the battery SOC, the constant current mode is used for the discharging process. The constant current and constant voltage mode is used for charging process. The charging process is turned off when the charging current drops to 0.1 C-rate at 4.2 V.

To evaluate the internal resistance during pulse heating, the direct current resistance (DCR) of the battery is adopted and calculated. The DCR consists of ohmic resistance and polarization resistance. The ohmic resistance causes an instantaneous voltage drop

at the discharging beginning. The polarization resistance induces the slow decline of the voltage in the following discharging process. As each discharging period is no longer than 1 s in this pulse heating scheme, a large part of the voltage drop is caused by the ohmic resistance while a small part is caused by the polarization resistance. The average DCR in the overall heating process is calculated as follows:

$$R_l = \frac{1}{N} \times \sum_i^N \frac{\Delta V_i}{I_i} \quad (2)$$

where R_l denotes the DCR; and ΔV_i and I_i refers to the voltage drop and current in i th discharging period, respectively. N is the total number of discharging periods. Then, the external circuit resistance R_E is calculated as follows:

$$R_E = \frac{1}{N} \times \sum_i^N \frac{U_{on,i}}{I_i} \quad (3)$$

where $U_{on,i}$ denotes the discharging voltage in i th discharging period.

The transient current in i th discharging period can be calculated as:

$$I_i = \frac{U_{off,i}}{R_{l,i} + R_{E,i}} \quad (4)$$

2.3. Definition of heating efficiency and energy consumption ratio

To evaluate the pulse heating method, the heating efficiency η_h shown in Eq. (5) is introduced. The heating efficiency is defined as the ratio between the energy consumption for battery temperature-rise and the total energy consumption for the heating process. The total energy consumption is composed of output power to external circuit, energy for temperature-rise and heat loss to the environment. The heating efficiency can be expressed as:

$$\eta_h = \frac{\int_{t_0}^{t_{end}} m C_p dT}{\int_{t_0}^{t_{end}} U_{on} I dt + \int_{t_0}^{t_{end}} m C_p dT + \int_{t_0}^{t_{end}} h A (T - T_A) dt} \quad (5)$$

where m is the mass of battery, C_p is the specific heat capacity of the battery and T is the temperature of the battery, h is the heat transfer coefficient at battery surface, A is the battery surface area, t_0 and t_{end} denote the initial and end state of heating. The heat transfer via battery surface is natural convection and heat transfer coefficient h is determined from the empirical correlation for the average Nusselt number in natural convection [27]:

$$Nu = \left\{ 0.825 + \frac{0.387 Ra^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (6)$$

where Nu is the average Nusselt number, Ra is the Rayleigh number and Pr is the Prandtl number. The average value for heat transfer coefficient is evaluated to be 5.32 W/m² K.

To evaluate the energy consumption of the heating process, the energy consumption ratio (ECR) is introduced and defined as the ratio of the energy consumed on heating process over total energy (TE) stored in the battery. The TE of the battery is 7 Wh which is the measured discharge power from 4.2 V to 2 V with 1 C-rate at 25 °C. Then, the ECR can be calculated as:

$$ECR = \frac{\int_{t_0}^{t_{end}} U_{on} I dt + \int_{t_0}^{t_{end}} m C_p dT + \int_{t_0}^{t_{end}} h A (T - T_A) dt}{TE} \quad (7)$$

2.4. Evaluation of battery capacity degradation

For low temperature battery heating, the life reduction and battery degradation are major concerns. To evaluate the impact of pulse heating on the battery degradation, three batteries are employed for life test. The battery-1 is heated up from −10 °C to 0 °C using pulse heating and the heating process is repeated for 100 times. The battery-2 is heated up from −10 °C to 0 °C using DC heating (9.5 A current) and the heating process is also repeated for 100 times. The battery-3 is the control group without applying heating process. After treated by different heating methods, the life test is then performed for three batteries. The life test is conducted using 1 C-rate discharge and 0.5 C-rate charge cycle. In the life test, the state-of-health (SOH) is employed to show the battery life. The SOH is the most commonly used indicator of battery degradation and can be calculated as [28]:

$$SOH = \frac{C_{dis,actual}}{C_{dis,initial}} \quad (8)$$

where $C_{dis,actual}$ is the actual discharge capacity of the battery (A h) and $C_{dis,initial}$ is the initial discharge capacity of the battery in each discharge/charge cycle. The battery life is defined as the cycle number at which the SOH drops to 0.8.

3. Results and discussion

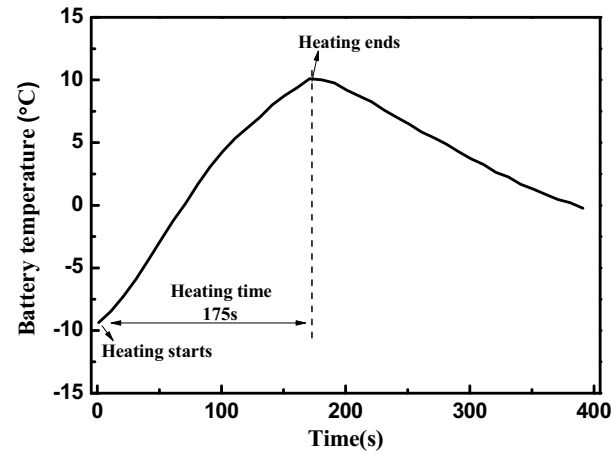
3.1. Temperature response in pulse self-heating

To acquire the temperature and voltage variation of the battery during self-heating, the pulse heating signal is applied to the battery. Heating is performed with the switching interval of 0.5 s. The initial ambient temperature is −10 °C, and heating is switched off when the battery reaches 10 °C. The SOC is set to 1. Fig. 3(a) shows the temperature evolution of the battery. The battery temperature increases rapidly once the self-heating is started. When heating is switched off at 175 s, the battery temperature remains 10 °C for around 10 s before a gradually decline. The delay of temperature decrement is caused by the heat conduction from battery core area to surface area.

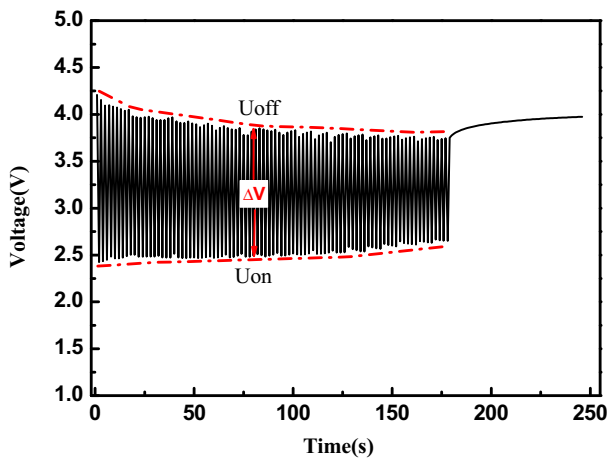
Fig. 3(b) depicts the battery voltage and heating current during heating. The voltage fluctuates with the fast periodic on-off operation during the self-heating process. The upper dash denotes U_{on} , and the lower dash line denotes U_{off} . The instantaneous voltage drop ΔV ($\Delta V = U_{on} - U_{off}$) occurs at the moments of each switching on operation. This voltage drop is induced by the internal resistance when the current passes through the battery. The initial voltage drop is approximately 1.8 V at the start, and its amplitude gradually shrinks when the self-heating proceeds. The battery gradually warms up during heating, resulting in decreased internal resistance. The amplitude voltage drop gradually decreases due to decreased internal resistance. When the heating is switched off, the battery voltage gradually resumes to a higher value around 4 V due to the absence of discharging current. The current in the circuit is presented in Fig. 3(c). The current curve during on-period of the circuit gradually drops when the heating proceeds. The rapid decline at the beginning is caused by the drop of U_{off} . The drop of the current slows down due to the decreasing internal resistance when the battery is warmed up. The maximum current is 11 A at the initial state and the minimum current is 8 A at the end of pulse discharge. The mean value of the current is calculated as 9.5 A.

3.2. Comparison of pulse heating and direct current heating

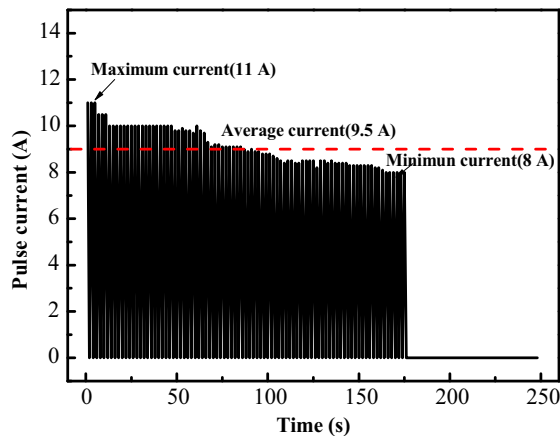
To show the effectiveness of pulse self-heating, the continuous DC heating is selected as the comparison reference. The testing



(a) Temperature evolution of the battery.



(b) Voltage pulse of the battery.

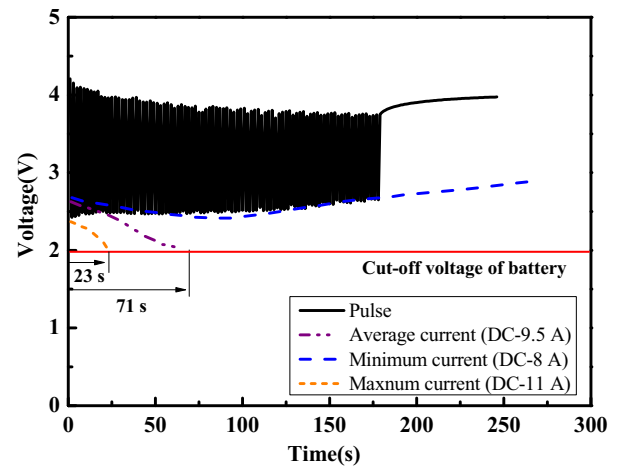


(c) Current pulse of the battery.

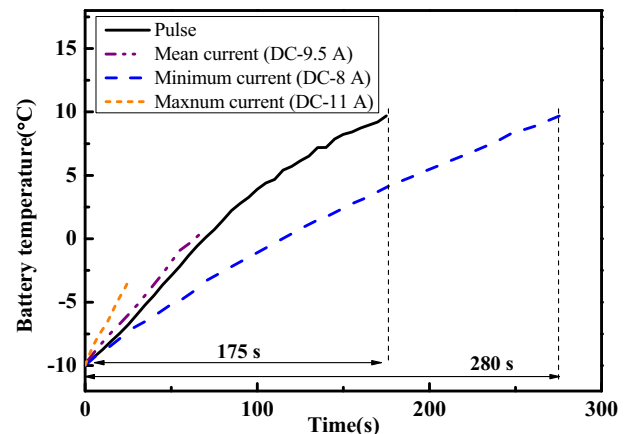
Fig. 3. Temperature rise of the three batteries with respect to time and the corresponding battery voltage and current change.

condition is the same with that shown in Fig. 3. Three currents values, namely maximum current (11 A), minimum current (8 A) and mean current (9.5 A), are selected to perform the continuous DC heating. During the test, the discharge operation is terminated when the battery temperature reaches 10 °C or the battery voltage drops to cut-off voltage (2 V). The battery temperature, voltage evolution, and energy consumption of pulse heating and continuous DC heating are obtained.

Fig. 4(a) shows the voltage profiles in DC heating and pulse heating. The voltage for DC heating with minimum current decreases at first and then gradually increases and the voltage remains larger than cut-off voltage for the entire heating process. The voltage decrement in the first half is caused by the polarization. The voltage increment in the second half is caused by the temperature increase of the battery, which reduces R_i . Limited by the cut-off voltage, the heating process keep only 71 s and 23 s for the DC heating with mean current and maximum current. As battery polarizations are more significant at mean current and maximum current, only voltage decrements are presented. For pulse heating, the U_{on} remains larger than cut-off voltage for entire heating process and the variation of U_{on} in the pulse heating approximates the discharge voltage in DC heating with minimum current. However, U_{on} is higher than the discharge voltage in DC heating with mean current, which indicates that the polarization resistance of pulse heating is lower than DC heating under similar discharge current. The lower polarization resistance in pulse heating is induced by the short discharge time in each on-period of pulse operation. As the discharge time (0.5 s) is much lower than the time constant (5–10 s) of concentration gradient voltage [29], the diffusion polarization is reduced and the higher discharge voltage is presented for the pulse heating. The comparison results indicate that the proposed pulse heating method increases the discharge capability of the battery under low temperature. With



(a) Battery voltage evolution.



(b) Battery temperature evolution.

Fig. 4. Comparison of pulse heating and continuous DC heating.

pulse heating, a higher discharge current can be achieved without worrying the over-discharge of the battery.

Fig. 4(b) shows the temperature evolutions for DC heating and pulse heating. For DC heating, the faster temperature increase can be obtained with higher discharge current. The peak temperature for DC heating with mean current and maximum current are 0.95 °C and −3.5 °C as the voltages for two DC heating processes drop to cut-off voltage before their battery temperatures reach the target temperature (10 °C). For the DC heating with minimum current, it takes 280 s to achieve the target temperature (10 °C) for the battery. In comparison, it takes 175 s to achieve the target temperature (10 °C) for the battery with pulse heating. To quantitatively evaluate two heating methods, heating efficiencies for two heating methods are calculated based on Eq. (5). The average temperature increase rate, and heating efficiencies are shown in Table 2. For DC heating, the temperature increase rate and the heating efficiency increase with the discharge current. The higher temperature increase rate implies the higher heat generation, which is caused by the higher polarization with higher discharge current. The higher heating efficiency is induced by the shorter heating time which decreases the convective heat loss to the environment. The temperature increase rate and heating efficiency of pulse heating are close to that of DC heating with mean current. In general, the pulse heating has shown advantages over DC heating in two aspects. Firstly, the pulse heating can provide much longer heating performance before the voltage drops to cut-off voltage when applying the same operation current. Secondly, the heating current in pulse heating can be designed higher than that in DC heating for the same discharge voltage limit, which will provide faster and safer heating process.

3.3. The impact of pulse heating on battery health

The SOH tests are performed to evaluate the battery degradation under two heating methods. The SOH evolutions for DC heating, pulse heating and fresh battery are compared. Fig. 5 shows the SOH variations for the three heating methods. The SOH in DC heating drops to 0.8 at 81 st cycle, while the SOH in pulse heating drops to 0.8 at 180th cycle. The SOH remains larger than 0.95 throughout the test for fresh battery. The battery degradation occurs both in DC heating and pulse heating according to reduced SOH. This is mainly caused by the large discharge current and consequent battery polarization during heating process. The battery life for pulse heating is 2.2 times of DC heating. This longer battery life in pulse heating is attributed to the higher discharge voltage during pulse process. The lower discharge voltage and higher polarization in DC heating easily cause cracks in the electrode material. Therefore, the capacity fade in DC heating is much severe than that in pulse heating.

3.4. Heating performance at different temperatures, switching intervals, and initial SOCs

3.4.1. Influence of switching interval

Fig. 6 shows the influence of the switching interval on heating duration and energy consumption ratio. The initial battery temperature is set to −10 °C, and the initial SOC is set to 1. Heating duration t_h denotes the time needed to heat the battery from −10 °C to

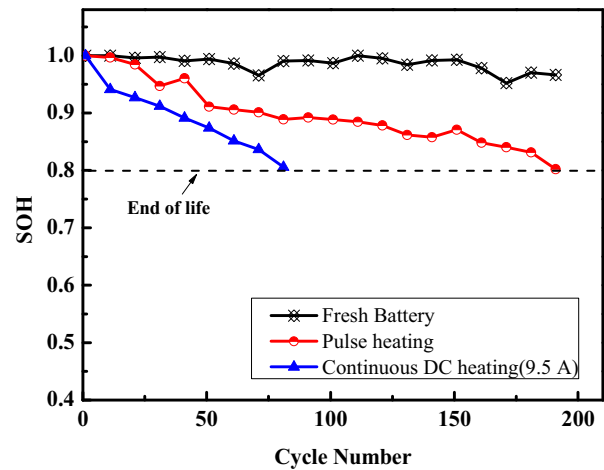


Fig. 5. Battery SOH under pulse heating and continuous DC heating.

10 °C. Heating duration and battery energy consumption only slightly increase with increased switching interval, whereas, the variations of heating duration and energy consumption are insignificant with the change of switching interval. This result implies that the heating performance is insensitive to the present switching time ranging from 0.1 s to 1 s. According to the energy balance, the battery energy consumption is in proportion to the temperature rise. Therefore, the energy consumptions at three different switching intervals are very close. The slightly higher energy consumption at switching interval of 1 s is caused by the higher heat release to the environment during longer heating duration.

3.4.2. Influence of ambient temperature

To investigate the ambient temperature effect on pulse heating, four different starting temperatures (0, −5, −10, and −20 °C) are set for tests. Fig. 7(a) shows the battery temperature evolutions at the four initial ambient temperatures. The battery temperatures at the four initial ambient temperatures increases linearly. When the temperature rise reaches 20 °C, the heating is switched off, and then the battery temperatures declines gradually and linearly. Heating durations are 163, 158, 175, and 230 s at the four ambient temperatures of 0, −5, −10, and −20 °C, respectively.

Fig. 7(b) depicts the average off-period voltage, internal resistance, and external circuit resistance variations with respect to ambient temperature in one test. The two parameters are highly temperature dependent. The voltage during off-period U_{off} increases significantly with increased ambient temperature. The increase of U_{off} is caused by the smaller polarization at higher temperature. In comparison, the internal resistance R_i decreases drastically with increased ambient temperature. The external circuit resistance R_E slightly increases with the increased temperature due to the positive temperature coefficient of copper wire resistance.

Heating durations and energy consumptions ratio at the four ambient temperatures are depicted in Fig. 7(c). The heating duration first decreases from −5 °C to −20 °C, and then it slightly increases from 0 °C to −5 °C. The shortest heating duration is obtained at the ambient temperature of −5 °C. Higher energy con-

Table 2
Heating efficiencies for DC heating and pulse heating.

Item	Pulse heating	DC heating (Mean current 9.5 A)	DC heating (Maximum current 11 A)	DC heating (Minimum current 8 A)
Temperature increase rate	6.74 °C/min	7.47 °C/min	15.8 °C/min	4.28 °C/min
Heating efficiency	27.4%	30.6%	38.2%	15.3%

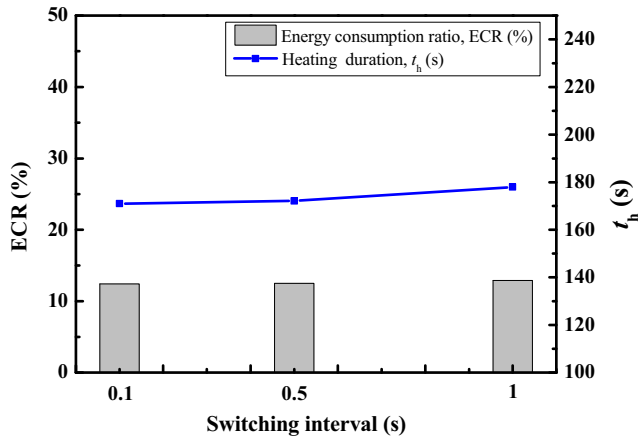


Fig. 6. Energy consumption and heating duration under different switching intervals.

sumption is required for longer heating duration. In essence, the heating duration is influenced by the competitive effect of U_{off} and R_i . The transient heat generation rate q_i within one on-period of pulse operation can be evaluated using U_{off} , R_i and R_E , which is formulated in Eq. (9):

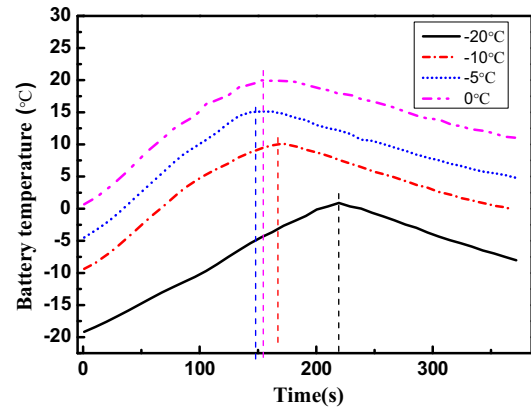
$$q_i = I_i^2 \times R_{i,i} = \left(\frac{U_{off,i}}{R_{i,i} + R_{E,i}} \right)^2 R_{i,i} = \frac{(U_{off,i})^2}{R_{i,i} + 2R_{E,i} + \frac{R_{E,i}^2}{R_{i,i}}} \quad (9)$$

According to Eq. (9), the increase in heat generation is determined by two factors: increasing the U_{off} and narrowing the gap between R_i and R_E . When the temperature increases from -20°C to -5°C , the U_{off} increases significantly, and the gap between R_i and R_E gradually increases as shown in Fig. 7(b). The influence of voltage is predominant at this stage, the heat generation is increased with the increasing of the U_{off} . As a result, the heating duration correspondingly decreases. When the temperature increases further from -5°C to 0°C , the gap between R_i and R_E becomes larger, whereas the increase in U_{off} becomes insignificant. The influence of R_i and R_E become dominant, and heating duration is slightly increased with decreased heat generation. In addition, the minimum energy consumption is observed at the ambient temperature of -5°C due to the shortest heating duration with the least heat dissipation to the environment.

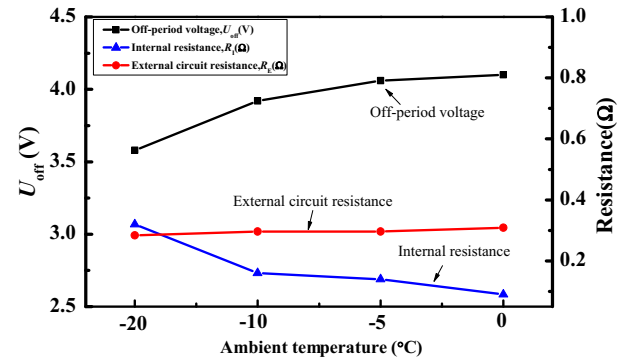
3.4.3. Influence of initial SOC

To investigate the effect of the initial SOC on the battery heating performance, battery self-heating tests are performed with the SOC varying from 0.2 to 1. The initial battery temperature is set to -10°C , and the switching interval time is 0.5 s. Fig. 8(a) presents the off-period voltage, internal resistance, and external circuit resistance variations with respect to SOC in one test. The U_{off} gradually increases with increasing SOC, which is induced by the increased chemical potential in cathode during lithium deintercalation. The R_i gradually decreases with the increased initial SOC due to the increased lithium diffusivity in the electrode material. The external circuit resistance remains unchanged with increasing SOC due to the constant temperature in the test.

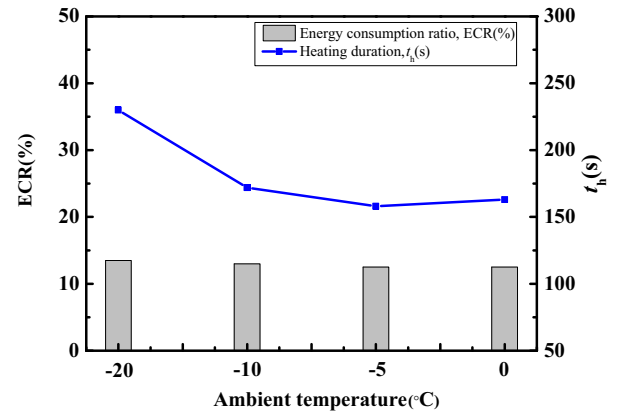
Fig. 8(b) shows the heating duration and energy consumption ratio with the SOC increasing from 0.2 to 1. The heating duration and the consumed battery capacity gradually drops with increased initial SOC. The reduction of heating duration can be explained by the coupling effect of U_{off} and R_i . The gap between R_i and R_E decreases first and then changes slightly. Meanwhile, the U_{off} increases significantly with the increased initial SOC as shown in



(a) Temperature evolution for the battery.



(b) Battery voltage and internal resistance at different ambient temperatures.



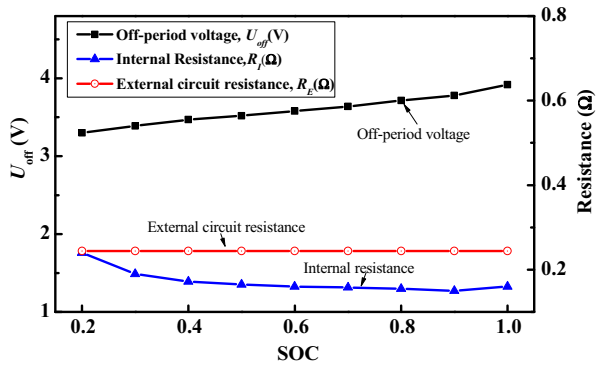
(c) Heating duration and energy consumption at different ambient temperatures

Fig. 7. Temperature rise at different ambient temperatures.

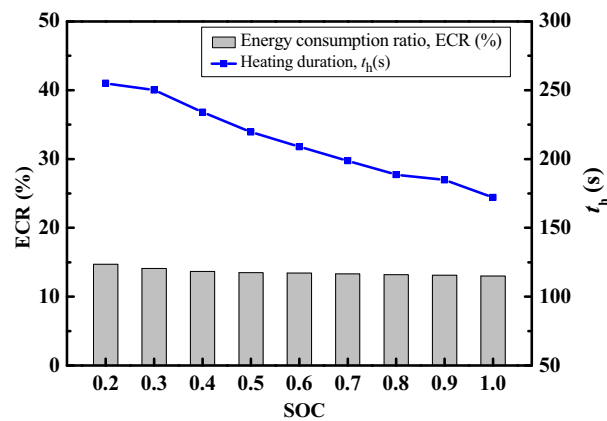
Fig. 8(a). The heating time decreases with increased SOC due to the increasing of the U_{off} . The reduced battery energy consumption is attributed to the less heat dissipation to the environment with the short heating duration. In real applications, the pulse heating is recommended to be performed at an SOC larger than 0.8 to achieve a heating duration less than 200 s.

4. Conclusions

A pulse internal self-heating strategy is proposed to achieve quick battery heating. An electric circuit is built to generate intermittently high current in the battery. Fluctuation of off-period voltage and on-period voltage are observed, and this fluctuation



(a) Battery voltage and internal resistance variation with SOC



(b) Heating duration and battery energy consumption

Fig. 8. Heating duration and energy consumption with respect to initial SOC.

amplitude gradually decreases as the heating proceeded. The heating performance of pulse heating is compared with conventional DC heating with the three operating currents, namely maximum current, minimum current and mean current, at the ambient temperature of -10°C . To achieve a 20°C temperature increment, the time consumption is 175 for pulse heating with the mean current of 9.5 A and the time consumption is 280 s for DC heating with 8 A constant current, respectively. The voltage of the battery drop to the cut-off voltage before the battery can be heated from -10 to 10°C when the current is increased to 9.5 A in DC heating. The temperature increase rate and the heating efficiency of pulse heating are close to that of DC heating with the mean current of 9.5 A. The shortest heating duration is obtained at -5°C , which is induced by the competitive effect of off-period voltage and internal resistance. Heating duration is insensitive to switching intervals ranging from 0.1 s to 1 s. Heating duration decreases with increased initial SOC, which results in an increase in the off-period voltage and a decrease in the internal resistance. An initial SOC higher than 0.8 is recommended to warm the battery with 20°C temperature rise within 200 s at an ambient temperature of -10°C . The application of pulse heating is practical, as less than 15% of the battery energy is consumed for a 20°C temperature increase. The present study is just the beginning of battery pulse heating. We pay more attention to the heating characteristics for pulse heating in this study. There is no doubt that the heating process is worthy especially when the exterior heating power is absent. The heating efficiency can be further increased by better control circuit and thermal insulation design.

Conflict of interest

The authors declared that there is no conflict of interest.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijheatmasstransfer.2019.02.020>.

References

- [1] S. Panchal, I. Dincer, M. Agelin-Chaab, R. Fraser, M. Fowler, Uneven temperature and voltage distributions due to rapid discharge rates and different boundary conditions for series-connected LiFePO₄ batteries, *Int. Commun. Heat Mass* 81 (2017) 210–217.
- [2] S. Panchal, I. Dincer, M. Agelin-Chaab, R. Fraser, M. Fowler, Transient electrochemical heat transfer modeling and experimental validation of a large sized LiFePO₄/graphite battery, *Int. J. Heat Mass Transfer* 109 (2017) 1239–1251.
- [3] M.S. Wu, K.H. Liu, Y.Y. Wang, C.C. Wan, Heat dissipation design for lithium-ion batteries, *J. Power Sources* 109 (2002) 160–166.
- [4] X.M. Xu, R. He, Review on the heat dissipation performance of battery pack with different structures and operation conditions, *Renew. Sust. Energy Rev.* 29 (2014) 301–315.
- [5] W.Q. Li, Z.G. Qu, Y.L. He, W.Q. Tao, Experimental and numerical studies on melting phase change heat transfer in open-cell metallic foams filled with paraffin, *Appl. Therm. Eng.* 37 (2012) 1–9.
- [6] W.Q. Li, Z.G. Qu, Y.L. He, Y.B. Tao, Experimental study of a passive thermal management system for high-powered lithium ion batteries using porous metal foam saturated with phase change materials, *J. Power Sources* 255 (2014) 9–15.
- [7] S.S. Zhang, K. Xu, T.R. Jow, The low temperature performance of Li-ion batteries, *J. Power Sources* 115 (1) (2003) 137–140.
- [8] H.C. Shiao, D. Chua, H.P. Lin, S. Slane, M. Salomon, Low temperature electrolytes for Li-ion PVDF cells, *J. Power Sources* 87 (1) (2000) 167–173.
- [9] S.S. Zhang, K. Xu, T.R. Jow, Charge and discharge characteristics of a commercial LiCoO₂-based 18650 Li-ion battery, *J. Power Sources* 160 (2) (2006) 1403–1409.
- [10] C.K. Huang, J.S. Sakamoto, J. Wolfenstine, S. Surampudi, The limits of low-temperature performance of Li-ion cells, *J. Electrochem. Soc.* 147 (8) (2000) 2893–2896.
- [11] S. Tippmann, D. Walper, L. Balboa, B. Spier, W.G. Bessler, Low-temperature charging of lithium-ion cells part I: Electrochemical modeling and experimental investigation of degradation behavior, *J. Power Sources* 252 (2014) 305–316.
- [12] J. Li, H. Tian, P. Wu, Researches on heating low-temperature lithium-ion power battery in electric vehicles, in: *IEEE Conference and Expo Transportation Electrification Asia-Pacific*, Beijing, China, 2014, pp. 1–6.
- [13] A. Pesaran, A. Vlahinos, T. Stuart, Cooling and preheating of batteries in hybrid electric vehicles, in: *6th ASME-JSME Thermal Engineering Joint Conference*, Hawaii, USA, 2003, pp. 1–7.
- [14] H.S. Song, J.B. Jeong, B.H. Lee, D.H. Shin, B.H. Kim, T.H. Kim, Experimental study on the effects of pre-heating a battery in a low-temperature environment, in: *IEEE Vehicle Power and Propulsion Conference*, Seoul, South Korea, 2012, pp. 1198–1201.
- [15] J. Zhang, H. Ge, Z. Li, Z. Ding, Internal heating of lithium-ion batteries using alternating current based on the heat generation model in frequency domain, *J. Power Sources* 273 (2015) 1030–1037.
- [16] H. Ruan, J. Jiang, B. Sun, W. Zhang, W. Gao, L.Y. Wang, Z.Y. Ma, A rapid low-temperature internal heating strategy with optimal frequency based on constant polarization voltage for lithium-ion batteries, *Appl. Energy* 177 (2016) 771–782.
- [17] J.G. Zhu, Z.C. Sun, X.Z. Wei, H.F. Dai, W.J. Gu, Experimental investigations of an AC pulse heating method for vehicular high power lithium-ion batteries at subzero temperatures, *J. Power Sources* 367 (2017) 145–157.
- [18] G. Ning, B. Haran, B.N. Popov, Capacity fade study of lithium-ion batteries cycled at high discharge rates, *J. Power Sources* 117 (2003) 160–169.
- [19] Y. Ji, C.Y. Wang, Heating strategies for Li-ion batteries operated from subzero temperatures, *Electrochim. Acta* 107 (3) (2013) 664–674.
- [20] M. Petzl, M. Kasper, M.A. Danzer, Lithium plating in a commercial lithium-ion battery—A low-temperature aging study, *J. Power Sources* 275 (2015) 799–807.
- [21] C.Y. Wang, G. Zhang, S. Ge, T. Xu, Y. Ji, X.G. Yang, Y.G. Leng, Lithium-ion battery structure that self-heats at low temperatures, *Nature* 529 (2016) 515–518.
- [22] X.G. Yang, G.S. Zhang, C.Y. Wang, Computational design and refinement of self-heating lithium ion batteries, *J. Power Sources* 328 (2016) 203–211.

- [23] X.G. Yang, T. Liu, C.Y. Wang, Innovative heating of large-size automotive Li-ion cells, *J. Power Sources* 342 (2017) 598–604.
- [24] Z.G. Lei, Y.W. Zhang, X.G. Lei, Improving temperature uniformity of a lithium-ion battery by intermittent heating method in cold climate, *Int. J. Heat Mass Transfer* 121 (2018) 275–281.
- [25] Z.G. Lei, Y.W. Zhang, X.G. Lei, Temperature uniformity of a heated lithium-ion battery cell in cold climate, *Appl. Therm. Eng.* 129 (2018) 148–154.
- [26] Y.Y. Chen, A. Sitzman, Testing a lithium ion battery as a pulsed power source, in: *IEEE Pulsed Power Conference*, 2005, pp. 485–488.
- [27] S.W. Churchill, H.H.S. Chu, Correlating equations for laminar and turbulent free convection from a vertical plate, *Int. J. Heat Mass Transfer* 18 (11) (1975) 1323–1329.
- [28] J. Jaguemont, L. Boulon, Y. Dubé, A comprehensive review of lithium-ion batteries used in hybrid and electric vehicles at cold temperatures, *Appl. Energy* 164 (2016) 99–114.
- [29] T. DuBeshter, J. Jorne, Pulse polarization for Li-ion battery under constant state of charge: Part I. Pulse discharge experiments, *J. Electrochem. Soc.* 164 (11) (2017) E3539–E3546.